2012 Q1

1. (a) $Q = m_s c_s \Delta T + m_s l_v$	1M+1M	
$= 0.02\,(2000)(110 - 100) + 0.02\,(2260000)$		
$= 400 + 45200$		
$= 45600$ J	1A	3
(b) $m_m c_m \Delta T_m = Q + m_s c_w \Delta T_w$	1M	
$0.2\,(3900)(T - 15) = 45600 + 0.02\,(4200)(100 - T)$		
$T = 76.0$ °C	1A	2
(c) The actual temperature of frothy milk is lower than the value calculated in (b).	1A	
Because energy loss by the steam is also gained by the surroundings	1A	
including the air/the metallic jug that holds the milk.		2

2019 Q1

1. (a) (i) $(1.5)(4200)(60 - 10) = m(3.34 \times 10^5) + m(4200)(10 - 0)$ $m = 0.837766$ kg ≈ 0.838 kg	1M+1M 1A 3	
(ii) More ice is needed to cool the container as it will release heat / thermal energy as well.	1A 1A 2	
(b) (i) Any ONE: Conduction: - Foam is a poor conductor (of thermal energy) and it minimizes the transfer of heat / thermal energy from the surroundings to the cool contents / ice cream (inside the bag). OR Radiation: - The shiny surface (inside) reduces the transfer of heat / thermal energy from the surroundings to the cool contents / ice cream (inside the bag) through emission of radiation OR Convection: - The zipper prevents convection between the hot air outside and the cool contents / ice cream (inside the bag).	1A 1	Note : Convection is the least effective transfer process in this case as the hot air outside is above the denser cool air around the ice cream.
(ii) (Radiation) Make the outer surface (of the bag) shiny.	1A 1	

2021 Q1

Solution	Marks	Remarks
1. (a) $Pt = ml$		
$l = \dfrac{150 \times (5\times60)}{0.016}$	1M	
$= 2812500\ \text{J kg}^{-1} \approx 2810\ \text{kJ kg}^{-1}$	1A	
	2	
(b) $C(100-22) = m_w c_w (22-20)$		
$C = \dfrac{0.100 \times 4200(22-20)}{(100-22)}$	1M	
$= 10.76923\ \text{J °C}^{-1} \approx 10.8\ \text{J °C}^{-1}$	1A	
	2	
(c) Since energy transferred by the boiling water is taken to be energy from the metal sphere / extra energy is transferred (by the boiling water) to the cup of water / the final temperature is higher.	1A	
The true value of C is smaller (than the calculated value).	1A	
	2	
(d) Not justified (with correct explanation).	1A	
Copper – (good) conductor while polystyrene – (good) insulator or poor conductor, more energy will be lost to the surroundings by conduction via a copper cup / most part of the polystyrene cup is still at room temperature, negligible energy is absorbed.	1A	
	2	

2024 Q1

1. (a) conduction / radiation	1A	
	1	
(b) Will not, as the silvery surface of aluminum foil is a poor radiation absorber / good radiation reflector.	1A+1A	
	2	
(c) (i) The water droplets come from the moisture / water vapour in the air (surroundings).	1A	
	1	
(ii) $E = (0.40 \times 10^{-3})(2.26 \times 10^{6})$	1M	
$= 904\ \text{J}$	1A	
	2	
2. (a) (i) $E_K = \frac{1}{2}\left(6.63\times10^{-26}\right)(500)^2$	1M	Accept: $(8.29 \sim 8.30) \times 10^{-21}$ J
$= 8.2875 \times 10^{-21}\ \text{J} \approx 8.29 \times 10^{-21}\ \text{J}$	1A	
	2	
(ii) $E_K = \dfrac{3RT_0}{2N_A}$		
$2\left(8.29\times10^{-21}\right)$		